

Linear algebra – 2

Vector spaces, dimension and rank

Homework

3 октября 2026 года

Theoretical minimum. The rank has four equivalent definitions — dimension of the column space, size of the largest non-zero minor, least inner dimension in a factorisation $A = BC$, least number of rank-one summands — and each of them is the convenient one in some problem. It is invariant under elementary operations and under multiplication by invertible matrices. The working inequalities are $\text{rank}(A + B) \leq \text{rank } A + \text{rank } B$, $\text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B)$, Sylvester's $\text{rank}(AB) \geq \text{rank } A + \text{rank } B - n$ and Frobenius' $\text{rank}(AB) + \text{rank}(BC) \leq \text{rank } B + \text{rank}(ABC)$. Everything else follows from $\dim \ker A + \text{rank } A = n$ and from the dimension formula for a sum of subspaces. In a space of functions, linear independence is proved by evaluating at well-chosen points or by differentiating; in a space of matrices, by pairing with a suitable matrix through the trace form.

1. Prove that every odd polynomial function of degree $2m - 1$ can be written as

$$P(x) = c_1 C_x^1 + c_2 C_{x+1}^3 + c_3 C_{x+2}^5 + \cdots + c_m C_{x+m-1}^{2m-1},$$

where $C_x^m = \frac{x(x-1)\cdots(x-m+1)}{m!}$.

2. Let n be a positive integer and let $P(x)$ be a polynomial of degree n with complex coefficients such that $P(0), P(1), \dots, P(n)$ are all integers. Prove that the polynomial $n!P(x)$ has integer coefficients.
3. (VJIMC 2002) Differentiable functions $f_1, \dots, f_n: \mathbb{R} \rightarrow \mathbb{R}$ are linearly independent. Prove that there exist at least $n - 1$ linearly independent functions among $f'_1, \dots, f'_n$.
4. (ISTCiM 2023) Let $n \geq 2$ and let X_n be the linear span of all $n!$

permutation matrices in the space $M_n(\mathbb{R})$ of real $n \times n$ matrices. Find $\dim X_n$.

5. (ФКН 2026, осень) Let V be a real vector space of dimension at least 3. Let $u \in V$ be a non-zero vector, and let $A: V \rightarrow V$ be a linear operator. Assume that $A(x) \in \text{span}\{x, u\}$ for every vector $x \in V$. Prove that there exists a scalar λ and a linear function $\varphi: V \rightarrow \mathbb{R}$ such that

$$A(x) = \lambda x + \varphi(x)u.$$

6. (VJIMC 2011) Let $n > k$ and let $A_1, \dots, A_k$ be real $n \times n$ matrices of rank $n - 1$. Prove that $A_1 A_2 \cdots A_k \neq O$.
7. (Romania national olympiad 2023) Let $A, B \in M_n(\mathbb{R})$. Show that $\text{rank } A = \text{rank } B$ if and only if there exist non-singular matrices $X, Y, Z \in M_n(\mathbb{R})$ with $AX + YB = AZB$.
8. (Putnam 1988) For a positive integer n let M_n be the $(2n+1) \times (2n+1)$ skew-symmetric matrix in which each entry in the first n subdiagonals below the main diagonal is 1 and each of the remaining entries below the main diagonal is -1 . Find $\text{rank } M_n$.
9. (Romania national olympiad 2016) Let $n \geq 2$ and let A, B, C be complex matrices such that A is invertible, B is obtained from A by replacing its first row with zeros, and C is obtained by putting the last $n - 1$ rows of A above a row of zeros. Prove that

a) $\text{rank}(A^{-1}B) = \text{rank}((A^{-1}B)^2) = \cdots = \text{rank}((A^{-1}B)^n);$

b) $\text{rank}(A^{-1}C) > \text{rank}((A^{-1}C)^2) > \cdots > \text{rank}((A^{-1}C)^n).$

10. (Proposed problems) Let $X, Y \in M_n(\mathbb{C})$ satisfy $Y^2 = YX - XY$ and $\text{rank}(X + Y) = 1$. Prove that $Y^3 = YXY = O_n$.

11. (Romania national olympiad 2004) Let $n \geq 2$.

a) Give an example of matrices $A, B \in M_n(\mathbb{C})$ with $\text{rank}(AB) - \text{rank}(BA) = \lfloor \frac{n}{2} \rfloor$.

b) Prove that $\text{rank}(XY) - \text{rank}(YX) \leq \lfloor \frac{n}{2} \rfloor$ for all $X, Y \in M_n(\mathbb{C})$.

12. (Romania national olympiad 2017) Let $n \geq 2$ and let $A, B \in M_n(\mathbb{C})$ satisfy $(AB)^3 = O_n$. Does it follow that $(BA)^3 = O_n$?
13. (ФКН, осень 2021) For $v, w \in \mathbb{R}^n$ let $v * w$ be the matrix whose (i, j) entry is $v_i w_j$. Assuming that v and w are linearly independent, find $\text{rank}(v * w - w * v)$.
14. (IMC) For $n \geq 2$, what are the minimal and the maximal possible ranks of an $n \times n$ matrix whose n^2 entries are precisely the numbers $1, 2, \dots, n^2$?
15. (Высшая лига 2020) Let $P, Q \in M_n(\mathbb{C})$ satisfy $P^2 = P$ and $Q^2 = Q$. Can $\text{rank}(E - P - Q)$ exceed $n - |\text{rank } P - \text{rank } Q|$?
16. (Proposed problems) Prove that $\text{rank}(A - ABA) - \text{rank}(B - BAB) = \text{rank } A - \text{rank } B$ for all $A, B \in M_n(\mathbb{C})$.